

消融实验结果

表 1: 消融实验结果：解质量对比（最优值加粗）

场景	配置	Best	Mean $\pm$ Std	Median
balanced ($N = M = 10$)	Vanilla DMDE	325672.23	325730.76 $\pm$ 101.38	325672.23
	+CR Control	325672.23	325789.29 $\pm$ 101.38	325847.82
	+PopInit	325672.23	325761.63 $\pm$ 154.85	325672.23
	Full LLM-DMDE	325672.23	325730.76 $\pm$ 101.38	325672.23
srp ($N = 10, M = 20$)	Vanilla DMDE	669221.05	682078.75 $\pm$ 11150.99	687912.37
	+CR Control	669221.05	682078.75 $\pm$ 11150.99	687912.37
	+PopInit	669221.05	682078.75 $\pm$ 11150.99	687912.37
	Full LLM-DMDE	669221.05	682078.75 $\pm$ 11150.99	687912.37

表 2: 计算时间对比（分口径）

场景	配置	总耗时 (s)	DMDE(s)	LLM 总耗时 (s)	LLM Init(s)	LLM CR(s)
balanced ($N = M = 10$)	Vanilla DMDE	59.2	59.2	—	—	—
	+CR Control	96.8	41.6	55.2	—	55.2
	+PopInit	99.4	59.1	40.2	40.2	—
	Full LLM-DMDE	166.6	41.2	125.4	60.4	65.1
srp ($N = 10, M = 20$)	Vanilla DMDE	48.0	48.0	—	—	—
	+CR Control	100.3	32.2	68.1	—	68.1
	+PopInit	120.5	48.2	72.3	72.3	—
	Full LLM-DMDE	171.2	32.5	138.7	66.6	72.1

协同效应分析

收敛速度对比

初始种群质量对比

表 3: 统计显著性检验 (Mann-Whitney U, A3 vs 基线)

场景	A3 vs A0	A3 vs A1	A3 vs A2
balanced ($N = M = 10$)	—	—	—
srp ($N = 10, M = 20$)	—	—	—

* $p < 0.05$, ** $p < 0.01$, n.s. 不显著

表 4: 协同效应分析: A3 增益 vs A1+A2 增益之和

场景	ΔA_1 (CR)	ΔA_2 (PopInit)	ΔA_3 (Full)	$\Delta A_1 + \Delta A_2$	协同比	结论
$S1$	-58.53	-30.87	+0.00	-89.40	-0.000	存在冗余
$S2$	+0.00	+0.00	+0.00	+0.00	n/a	无显著差异

表 5: 收敛速度对比: 达到 X% 改进所需的代数 (cost 场景下取中位数)

场景	配置	达到 90% 改进	达到 95% 改进	达到 99% 改进
balanced ($N = M = 10$)	Vanilla DMDE	152	152	240
	+CR Control	70	140	140
	+PopInit	220	230	270
	Full LLM-DMDE	100	270	460
srp ($N = 10, M = 20$)	Vanilla DMDE	—	—	—
	+CR Control	—	—	—
	+PopInit	—	—	—
	Full LLM-DMDE	—	—	—

表 6: 初始种群质量对比 (PopInit 效果验证; A1+CR Control 不影响初始种群, 故省略)

场景	配置	初始 Mean $\pm$ Std	初始 Median
balanced ($N = M = 10$)	Vanilla DMDE	465205.83 $\pm$ 8362.75	467620.05
	+PopInit	456937.94 $\pm$ 7779.32	459584.54
	Full LLM-DMDE	455851.13 $\pm$ 5175.79	457157.33
srp ($N = 10, M = 20$)	Vanilla DMDE	859931.52 $\pm$ 19609.41	869832.94
	+PopInit	859931.52 $\pm$ 19609.41	869832.94
	Full LLM-DMDE	859931.52 $\pm$ 19609.41	869832.94